Identical matter in local dS_3: Bose normalization and a short-time witness

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Abstract

We construct the two-identical-particle sector of the previously specified zero-gravity observer model by descending the matter permutation through group averaging. Canonical Fock normalization gives a direct term plus an exchange term, rather than a universal factor of two. The exchange matrix element is the existing physical scalar Racah element within the full channel space. The regulated two-creation order difference vanishes exactly at coincident times. A spherical test proves that its quadratic small-time coefficient is strictly positive for the compact real packet considered here. Raw quadrature and precision failures are retained; no finite-gravity or complete-Fock construction is claimed.

1 Model and labelled Hilbert space

Use radius $\ell = 1$, Newton constant $G_N = 0$, proper $SO^+(3,1)$, two distinguishable scalar BD observers with mass labels at least $s > 0$, and a single free KG-normalized scalar matter species $p > 0$. Couplings are one, $\rho(c) = 2c^2/\pi$, and $h_{\mathrm{m}}(c) = \sqrt{1+c^2}$. Keep the same real nonnegative initial packet:

$$\rho(c)|f(c)|^2 = \frac{630}{0.4}x^4(1-x)^4\mathbf{1}_{[0.4,0.8]}(c), \qquad x = (c-0.4)/0.4.$$

It has unit norm and its zero extension is C^1. Numerical examples take $s = 0.2$, $p = 0.6$. Every creation block carries the already defined half-period regulator $e^{\pi H_{\mathrm{out}}/2}Ae^{-\pi H_{\mathrm{in}}/2}$. At real time t the convention is $A(t) = e^{-iH_{\mathrm{out}}t}Ae^{iH_{\mathrm{in}}t}$. These are creation blocks, not self-adjoint unsmeared fields.

First temporarily label the two equal-mass matter slots. In the completed physical four-body space $\mathcal{H}_{\mathrm{lab}}$ the slot permutation F obeys

$$\mathsf{F}^\dagger = \mathsf{F}, \qquad \mathsf{F}^2 = 1, \qquad P_+ = (1+\mathsf{F})/2.$$

Indeed, on finite-K, compact-mass tests it commutes with the diagonal group action. The convergent four-body group average therefore satisfies $\eta(\mathsf{F}u, \mathsf{F}v) = \eta(u,v)$ and $\eta(u, \mathsf{F}v) = \eta(\mathsf{F}u, v)$. It preserves the null space, descends to the quotient, and extends by continuity as a unitary involution. The Bose space is $P_+\mathcal{H}_{\mathrm{lab}}$. This argument concerns the full four-body space, including all internal spins, rather than the scalar subblock in a selected pairing.

2 Fock normalization and the exchange form

For orthonormal matter modes, let $M_j(t)$ be the observer operator multiplying $a_j^\dagger$. On a finite-mode core a labelled chain has coefficients

$$C_{jk} = M_j(t_2)M_k(t_1)f.$$

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Direct use of $a_j^\dagger|n_j\rangle = \sqrt{n_j+1}|n_j+1\rangle$ gives the symmetric tensor

$$B(t_2,t_1) = \sqrt{2}\,P_+C = \frac{C+\mathsf{F}C}{\sqrt{2}}. \tag{1}$$

The factor $\sqrt{2}$ is the usual Fock factor [1]. It is not an additional choice of rigging-map or Haar normalization. The group-average argument above and the bounded regulated labelled $1\to 2$ block extend (1) to the physical completion. In particular $\|A^+_{B,1\to2}\| \le \sqrt{2}\|A^+_{\mathrm{lab},1\to2}\|$. This does not construct the whole Fock tower or a common self-adjoint field domain.

For any two labelled chain states k, l,

$$\langle B_k, B_l\rangle = \langle k,l\rangle + \langle k,\mathsf{F}l\rangle, \qquad 0 \le \|B_k\|^2 \le 2\|k\|^2. \tag{2}$$

The exchange expectation may have either sign for general k. Equality to $2\|k\|^2$ requires $\mathsf{F}k = k$. Noncommuting observer coefficients do not ensure this. Setting two distinguishable species labels equal only constructs $\mathcal{H}_{\mathrm{lab}}$; it does not apply the Bose projection.

Use the previous $0000 = 1$ scalar intertwiners J_b in the pairing $(a,p)|(c,p)$ and

$$W_b = \rho(b)K_{abp}K_{bcp}, \qquad e_b = \sqrt{W_b}J_b.$$

The flip maps J_d to the opposite pairing, with the same spherical normalization. Thus

$$U_{00}(b,d;a,p,c,p) = \langle e_b, \mathsf{F}e_d\rangle.$$

This is Hermitian as a kernel in b, d. Its full-space origin implies contraction bounds; one must not require its scalar restriction to square to one. The physical normalization and shadow phases are those fixed in the earlier LPRS dictionary [2]. No further factorial or shadow-reflection factor is inserted. The direct/crossed decomposition also has clear precedents in the two-dimensional observer calculation [3].

3 Two times and a stable order difference

Define the real base amplitude, supported at $a, b, c \ge s$,

$$T_b(a,c) = \frac{f(c)E_{abp}E_{bcp}}{\sqrt{W_b}}\,e^{\pi[h_{\mathrm{m}}(a)-h_{\mathrm{m}}(c)]/2}, \qquad \alpha_b = e^{-it[h_{\mathrm{m}}(a)-h_{\mathrm{m}}(b)]}, \quad \beta_b = e^{-it[h_{\mathrm{m}}(b)-h_{\mathrm{m}}(c)]}.$$

Let $k = \int T_b\alpha_b e_b\,\mathrm{d}b$ and $l = \int T_b\beta_b e_b\,\mathrm{d}b$, with the outer direct integrals in a, c understood. The physical Bose states are $B_{t0} = \sqrt{2}P_+k$ and $B_{0t} = \sqrt{2}P_+l$. Put

$$N = \langle k,k\rangle = \langle l,l\rangle, \quad Y = \langle k,l\rangle, \quad E_1 = \langle k,\mathsf{F}k\rangle, \quad E_2 = \langle l,\mathsf{F}l\rangle, \quad Z = \langle k,\mathsf{F}l\rangle.$$

Then

$$\|B_{t0}\|^2 = N + E_1, \qquad \|B_{0t}\|^2 = N + E_2, \tag{3}$$

$$\langle B_{t0}, B_{0t}\rangle = Y + Z, \qquad D_B(t) = 2N + E_1 + E_2 - 2\,\mathrm{Re}(Y+Z). \tag{4}$$

All exchange terms use the same unsymmetrized, complex U_{00}. For example their common weight is

$$\pi_0(c)\frac{g(a)}{g(c)}\sqrt{\widehat{R}_p(a,b)\widehat{R}_p(b,c)\widehat{R}_p(a,d)\widehat{R}_p(d,c)}\,U_{00}(b,d;a,p,c,p),$$

and the phases for E_1, E_2, Z are respectively

$$e^{it[h_{\mathrm{m}}(d)-h_{\mathrm{m}}(b)]}, \quad e^{it[h_{\mathrm{m}}(b)-h_{\mathrm{m}}(d)]}, \quad e^{it[h_{\mathrm{m}}(a)+h_{\mathrm{m}}(c)-h_{\mathrm{m}}(b)-h_{\mathrm{m}}(d)]}.$$

The direct terms use the positive density $\pi_0(c)g(a)\widehat{R}_p(a,b)\widehat{R}_p(b,c)/g(c)$. Here

$$g(x)=\frac{xe^{\pi/(\sqrt{1+x^2}+x)}}{1+e^{-2\pi x}},\quad \widehat{R}_p(a,b)=\frac{A-B}{8\pi(A+\cosh\pi p)(B+\cosh\pi p)},$$

with $A=\cosh\pi(a+b)$ and $B=\cosh\pi(a-b)$.

For computation set $e_b^{\mathrm{en}}=2h_{\mathrm{m}}(b)-h_{\mathrm{m}}(a)-h_{\mathrm{m}}(c)$ and

$$\delta_b=\alpha_b-\beta_b=2ie^{-it[h_{\mathrm{m}}(a)-h_{\mathrm{m}}(c)]/2}\sin(te_b^{\mathrm{en}}/2).$$

The stable formula is

$$D_B=\|\delta T\|^2+\langle\delta T,\mathsf{F}\,\delta T\rangle. \tag{5}$$

It gives $D_B(0)=0$ algebraically, without subtracting large nearly equal overlaps. Equal masses alone do not justify deleting the exchange term. A four-*field* vacuum correlator would also include the intermediate vacuum contraction; it is not computed by this two-creation sector.

4 A strict small-time result

All polynomial mass moments of T are finite. For instance the compact initial support, $g(a)\le a+C_0$, and $\widehat{R}_p(a,b)\le e^{-\pi|a-b|}/(4\pi)$ give an exponentially integrable majorant in the two mass differences. Consequently $j_b=e_b^{\mathrm{en}}T_b$ is in the labelled physical space. Multiplication by the common outer phase in δ_b is unitary and commutes with F. Using $|\sin y-y|\le|y|^3/6$ gives

$$D_B(t)=t^2Q_B+O(t^4),\qquad Q_B=\|j\|^2+\langle j,\mathsf{F}j\rangle=\|\sqrt{2}P_+j\|^2\ge 0. \tag{6}$$

This follows in Hilbert norm, with the sixth mass moment controlling the remainder. Equation (5) is even in real t.

We now show $Q_B>0$ for the stated packet. The exact local vertex from the preceding note is

$$E_{abp}=C_p\sqrt{\frac{\cosh\pi b}{\cosh\pi a}}H_p(a,b),\quad C_p=\frac{1}{4\sqrt{\pi}}\sqrt{\frac{\sinh\pi p}{p}}.$$

$$H_p(a,b)=r_p(a-b)-r_p(a+b),\qquad r_p(x)=\frac{1}{\cosh\pi x+\cosh\pi p}.$$

The all-spherical component is invariant under the matter flip. The derivative of the Bose order difference at zero has this component equal to $i\sqrt{2}f(c)$ times a positive common leg factor times

$$J(a,c)=\int_s^\infty H_p(a,b)H_p(b,c)[2h_{\mathrm{m}}(b)-h_{\mathrm{m}}(a)-h_{\mathrm{m}}(c)]\,\mathrm{d}b.$$

For $a=c>s$ this becomes $2\int_s^\infty H_p(c,b)^2[h_{\mathrm{m}}(b)-h_{\mathrm{m}}(c)]\,\mathrm{d}b$. Pair $b=c-x$ with $b=c+x$, for $0<x<c-s$. Since r_p strictly decreases on the positive axis,

$$H_p(c,c+x)>H_p(c,c-x)>0.$$

Strict convexity and monotonicity of h_{m} also imply

$$h_{\mathrm{m}}(c+x)-h_{\mathrm{m}}(c)>h_{\mathrm{m}}(c)-h_{\mathrm{m}}(c-x)>0.$$

The paired contribution is therefore strictly positive, and the remaining $b>2c-s$ contribution is positive. Thus $J(c,c)>0$.

Continuity, with exponential domination, implies positivity in some uniform strip $|a-c|<\varepsilon$ for c in any compact interval $I\subset(0.4,0.8)$, with ε sufficiently small to keep $a>s$. Choose an all-spherical compact-mass test with coefficient $f(c)\mathbf{1}_I(c)\mathbf{1}_{|a-c|<\varepsilon}$. Its four-body group-average norm is finite, and its pairing with $\sqrt{2}P_+j$ is strictly positive. Physical Cauchy–Schwarz proves $Q_B>0$. Hence $D_B(t)>0$ for sufficiently small nonzero t. The proof establishes existence of the strip and time neighborhood; no numerical size for either is certified. This effect is a finite-mass observer result at $G_N=0$, not a gravitational chaos exponent.

5 Mass tails and a free benchmark

Let T_L restrict $a, b \le L$ and retain the whole initial packet. The preceding compact-packet bounds apply at $q = p$:

$$\|T - T_L\|^2 \le \epsilon_L, \qquad \|T\|^2 \le N_{\max}.$$

Time phases do not change these norms. The direct inner-product tail is at most ϵ_L, since the cutoff commutes with the phase multipliers. The exchange tail is at most $2\sqrt{\epsilon_L N_{\max}}$. For (5), the direct tail is at most $4\epsilon_L$ and the exchange tail at most $8\sqrt{\epsilon_L N_{\max}}$. Thus

$$|D_B - D_{B,L}| \le 4\epsilon_L + 8\sqrt{\epsilon_L N_{\max}}. \tag{7}$$

This bound is not optimized for $t \to 0$ and does not certify tiny numerical differences at short times. For $L = 12$, $\epsilon_L \le 1.16353\,10^{-16}$, $N_{\max} \le 1.36456\,10^{-3}$, and the right side is $3.18767\,10^{-9}$. The internal physical spectrum remains infinite.

In a fixed-observer free benchmark, let u_t be a one-particle vector with $\langle u_t, u_0\rangle = L_p(t)$, where

$$L_p(t) = \frac{\sin(pt)}{4\pi \sinh(\pi p)\sinh t}, \qquad L_p(0) = \frac{p}{4\pi \sinh(\pi p)}.$$

The Fock identity gives

$$\|a^\dagger(u_t)a^\dagger(u_0)\Omega\|^2 = L_p(0)^2 + |L_p(t)|^2, \qquad D_{\text{free}}(t) = 0.$$

At $t = 0$ the norm is $2L_p(0)^2$. This is the required free target. We have not yet established convergence of the complete Bose exchange term to it as observer mass tends to infinity; the earlier noncrossed limit alone is insufficient.

6 Numerics, independent checks, and limitations

Three full complex-kernel grids retain 1,216,480 samples in NPZ files. The final grid uses $L = 12$, six Gauss nodes per mass segment, twelve initial-packet nodes, and the existing $(N, K) = (128, 12)$ special-function series settings. The final raw values, with digits shown for comparison rather than accuracy certification, are

t	$\|B_{t0}\|^2$	$D_B(t)$
0	$2.23450375\,10^{-4}$	0
0.001	$2.23450359\,10^{-4}$	$3.17557610\,10^{-11}$
0.01	$2.23448793\,10^{-4}$	$3.17553681\,10^{-9}$
0.5	$2.19598612\,10^{-4}$	$7.70115634\,10^{-6}$
1	$2.09164392\,10^{-4}$	$2.83756540\,10^{-5}$
2	$1.79403681\,10^{-4}$	$8.95584228\,10^{-5}$

The raw moment estimate is $Q_B \simeq 3.17557649\,10^{-5}$. Holding $L = 12$, the initial nodes, and the series settings fixed, mass-node refinement from four to six changes $D_B(2)$ by $2.26752\,10^{-8}$ and the full Bose overlap by at most $1.64309\,10^{-8}$ over these times. The coarser-to-finer comparison is empirical. In particular it exceeds the analytic mass-tail bound; that bound is not a total error bar.

All raw grids have been replayed from the unsimplified local vertices E, inverse norms W, and density ρ, rather than the simplified $\widehat{R}$ expression. The largest replay difference is below $3\,10^{-17}$. The final integrated absolute flip-Hermiticity residual is $6.78\,10^{-18}$ or smaller; this is a diagnostic, not a bound on real symmetric numerical error. Six selected large weighted-condition proxies have 55-digit reference comparisons, with maximum observed relative error about $7.16\,10^{-11}$. Neither sample selection nor condition numbers certify all remaining nodes.

The occupation-basis tests independently construct finite Fock matrices with three observer levels and four matter modes, including noncommuting observer coefficients. A separate finite cyclic-group average tests permutation descent algebraically; it is not a numerical replacement for noncompact de Sitter group averaging. High-precision one-dimensional integrals check the positive diagonal pairing argument.

A sample initially intended as regular, $(a, b, c, d) = (0.7, 0.5, 0.6, 0.8)$, lies extremely close to $a + c = b + d$ at binary64 precision. The old vectorized formula loses about 160% relative accuracy there and fails flip Hermiticity. The original failed diagnostic is retained. Selected-point reevaluation at 55 and 75 decimal digits restores consistency; it does not certify the whole vectorized grid. Another retained far-tail point has roughly 0.128% relative error. No raw kernel is silently made real, symmetrized, or replaced. The scientific proof of (6) does not depend on this special-function numerical representation.

Next work must establish stable large-mass crossed limits and their free Bose benchmark, then address the full Fock domain and finite gravity. The present note does not complete de Sitter holography.

References

[1] J. Dereziński and C. Gérard, *Asymptotic completeness in quantum field theory. Massive Pauli–Fierz Hamiltonians*, June 1997 manuscript, Secs. 2.2–2.3, author-hosted manuscript.

[2] J. Liu, E. Perlmutter, V. Rosenhaus and D. Simmons-Duffin, *d-dimensional SYK, AdS Loops, and* $6j$ *Symbols*, arXiv:1808.00612, v3.

[3] D. K. Kolchmeyer and H. Liu, *Chaos and the Emergence of the Cosmological Horizon*, arXiv:2411.08090, v1, Sec. 6.3.